Building Change Detection for VHR Remote Sensing Images via Local–Global Pyramid Network and Cross-Task Transfer Learning Strategy

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***Abstract*— Building change detection (BCD) for very-high-spatial-resolution (VHR) remote sensing images is very important and challenging in the field of remote sensing, as the building is one of the most significant and valuable man-made ground targets. This article proposes a local–global pyramid network (LGPNet) that combines a local feature pyramid module (LFPM) and a global spatial pyramid module (GSPM) for various building feature extraction. The LFPM is constructed using the convolutional kernel with three different pyramid scales, and then, the local pyramid features are obtained by adding features of each scale. In the GSPM, the global spatial pyramid features are extracted by adaptive average pooling to acquire global contextual information from different fields of view on deep features. The LFPM and the GSPM work in a parallel and complementary manner to capture discriminative features of various buildings. In addition to the LFPM and the GSPM, the proposed LGPNet also employs two general attention mechanisms, i.e., the position attention module and the channel attention module, which can select and emphasize adaptively some building features with high semantic responses. Besides, in order to mitigate the influence of other ground targets to a certain extent, a cross-task transfer learning strategy is introduced to make the LGPNet focus on the building, which significantly improves the performance of our method. Extensive experiments on two public available BCD datasets show that the proposed LGPNet can achieve significant improvement compared with eight other state-of-the-art methods. The source code and the pretrained model will be released at https://github.com/TongfeiLiu/LGPNet.**

***Index Terms*— Attention mechanism, building change detection (BCD), pyramid network, transfer learning, very-high-spatial-resolution (VHR) remote sensing images.**

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I. Introduction

CHANGE detection (CD) is to acquire changed information in two or more periods of images at the same geographical location [1], [2]. With a large number of high-spatial-resolution (HR) and very-HR (VHR) remote sensing images captured, there exists a great opportunity to observe more detailed ground targets, especially buildings, which are the most important and meaningful products of human modification on the Earth's surface. In this context, building CD (BCD) has received continuous attention and is utilized in many applications, such as urban environment [3], urban development and planning [4], and disaster assessment [5], [6].

During the last decade, with the development of CD techniques, many studies have been made and applied to BCD with VHR remote sensing images [7]–[9]. These approaches can be mainly divided into two categories, i.e., the postdetection comparison methods and the direct CD methods. The first approaches first detect the building maps from bitemporal images, respectively, and then acquire the changed buildings by comparing building maps. However, it is very challenging to obtain accurate building maps from bitemporal HR or VHR remote sensing images, which may lead to the accumulation of errors [10], [11]. The second methods accomplish BCD by measuring change magnitude with designed or generated handcraft features. For example, Huang *et al.* [12] developed a BCD method based on morphological building index for HR remote sensing images; Tian *et al.* [7] applied stereo imagery and digital surface models by integrating height changes and the Kullback–Leibler divergence for BCD. In addition, some general CD methods can be utilized for extracting the building changes [13]–[15]. Although these approaches based on handcraft features can acquire BCD results, there remains great difficult to apply these approaches to complicated scenes, and the accuracy is always limited.

In recent years, deep learning-based methods, especially deep convolutional neural networks (CNNs), have attracted great attention in the field of remote sensing [16]–[18], such as building and road extraction [19]–[21], hyperspectral classification [22], and remote sensing image denoising and

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superresolution [23], [24]. Moreover, many studies have applied deep learning-based techniques to CD on remote sensing images and improved the detection accuracy by training a CNN to excavate deep semantic information [25], [26]. In the early stage, Gong *et al.* [27] proposed a CD framework for synthetic aperture radar (SAR) images by training a deep neural network, which can omit calculating difference images to detect CD maps directly; Gao *et al.* [66] exploited principal component analysis network for automatic CD of SAR images. These studies have demonstrated that deep learning-based methods have achieved improvements in CD performance compared with traditional methods to a certain extent.

As a fully convolutional network (FCN) is successfully applied in natural image semantic segmentation [28], many scholars have regarded some tasks, such as classification, object detection, and CD of remote sensing images as semantic segmentation tasks, i.e., remote sensing image classification can be treated as multiobjective semantic segmentation [29], while object detection or CD can be treated as single-objective semantic segmentation [30], [31]. Based on this, many researchers have made sustained efforts on CD with the FCN, particularly the semantic segmentation network represented by U-Net [32], which has been widely used in CD. Lv *et al.* [33] employed a dual-path FCN based on U-Net by end-to-end training for landslide inventory mapping with VHR images. In the literature [34], an end-to-end CD approach on the basis of improved U-Net++ was proposed for HR satellite images. Three end-to-end architectures, namely, fully convolutional early fusion (FC_EF), fully convolutional Siamese-concatenation (FC_Siam_Conc), and fully convolutional Siamese-difference (FC_Siam_Diff), were proposed based on U-Net by the concatenation or Siamese strategy for CD [35]. Other references can be found in [36] and [37]. These approaches further promote the development of deep learning techniques in CD of remote sensing images.

Despite that the aforementioned methods have some advantages, these methods are unreliable for BCD. That is because BCD only focuses on buildings that are susceptible to the infestation of complicated backgrounds in dense urban scenes. Recently, some studies have been developed for BCD with VHR images. For example, Ji *et al.* [38] proposed a building instance CD framework, which contains a building extraction network and an FCN to obtain a building change map (BCM). A dual-task constrained Siamese CNN was designed for BCD on VHR aerial remote sensing images in the literature [10]. Saha *et al.* [39] transcoded image domain between SAR and optical images by exploiting cycle-consistent generative adversarial network (CycleGAN), and then, the deep change vector analysis and fuzzy rules were used to identify changed buildings. A multitask learning model based on the encoder–decoder architecture was proposed for fine-grained BCD, which can acquire "from-to" changes [40]. The above approaches can train the end-to-end model to capture deep representative features, so as to provide pixelwise prediction results. However, these models are short of adapting buildings of various scales and shapes.

Recently, some studies have probed solutions for learning the representative building features of different scales to overcome the aforementioned limitation. Sun *et al.* [40] opened an aerial and satellite building extraction and BCD datasets, namely, WHU, and devised a Siamese U-Net (SiU-Net) for building extraction. This model consists of two branches with a weight-shared strategy, and the original image and their downsampling counterparts are used as the input of two branches, respectively. It can provide good performance for BCD, especially for large buildings, as the downsampling counterparts of the original image enlarge the receptive field, which improves the model's ability to adapt buildings of different scales. Besides, attention mechanism is becoming more popular in CD applications. Ji *et al.* [41] promoted dual-attentive fully convolutional Siamese networks that adopt a dual-attention mechanism to capture long-range dependencies for obtaining discriminative feature representations. In addition to the attention mechanism, the model based on a pyramid structure is helpful to enhance the ability of feature representations for various objects. An attention-guided Siamese network based on pyramid feature was designed for BCD with orthoimagery, and a global coattention mechanism was introduced to consider the correlation of the feature pairs [11]. Chen *et al.* [43] constructed a spatial–temporal attention-based network (STANet), which divides the image into multiscale subregions, and the self-attention is introduced to each subregion in view of the object of different scales. These methods can accomplish a pixelwise BCM directly, but it still has the following limitations for BCD.

1) The detected BCM is often difficult to obtain a completely changed building instance, especially when the shapes and scales are very discrepant and irregular, which may lead to a poor visual result. As shown in Fig. 1, the small building depicted by the yellow

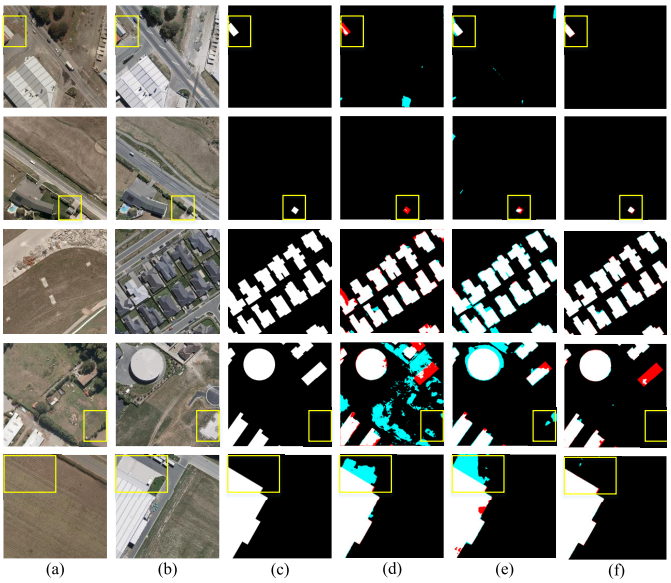


Fig. 1. Example results of the proposed LGPNet, SiU-Net [41], and STANet [42] on the WHUCD dataset [41]: (a) T_1-time images, (b) T_2-time images, (c) GT map, (d) SiU-Net [41], (e) STANet [42], and (f) proposed LGPNet. (Notations: red, cyan, white, and black denote missed detection pixels, false detection pixels, correct detection changed pixels, and correct detection unchanged pixels, respectively.)

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rectangular in the second row is obviously incomplete in the BCMs of SiU-Net [41] and STANet [42].

2) Since the spectra and structure of other ground targets may be similar with buildings in the complex urban scene, it is likely to acquire false detections in BCMs. As illustrated in the fourth and fifth rows with the yellow rectangle of Fig. 1, there are many false detected pixels in the BCMs of SiU-Net [41] and STANet [42].

The aforementioned drawbacks are mainly due to two reasons. The first reason is that, as illustrated in Fig. 1, multiscale (including buildings with large, small, and general scales) and multishape (including square, rectangle, round, and other irregular shapes) buildings usually exist in dense urban areas. These models cannot simultaneously excavate representative features of multiscale and multishape buildings, which affects the performance of these models. The second reason is that, as shown in Fig. 1, various ground objects are typically contained in urban scenes, such as trees, grass, roads, vehicles, and bridges. In other words, for an urban scene, the nonbuilding areas identified as "background" always involve a far more complicated and heterogeneous object, which may lead to more false detected pixels as the captured features are insufficiently discriminative. Therefore, it is necessary to find effective strategies to make the model focus on buildings to weaken the false detection triggered by other ground targets.

To overcome these limitations, we propose a local–global pyramid network (LGPNet) for BCD of bitemporal VHR remote sensing images in this article. The motivations of this approach lie in two aspects. For one thing, the local reception field is neither sufficient to identify the large buildings nor to capture the relationship between the buildings and other objects over the whole image to provide a precise location of buildings. Therefore, introducing global features is helpful to mitigate the limitation of using local features only. For another thing, the global features may implicate a much more complex background in the dense urban scene, which may lead to inaccurate and unreliable features. Hence, introducing local features can alleviate the unexpected interference triggered by global features. Based on the above motivations, a local–global pyramid building feature extractor is devised in the proposed LGPNet. This feature extractor is constructed with a local feature pyramid module (LFPM) and a global spatial pyramid module (GSPM) to capture the representative features from local and global perspectives, respectively. Besides, two general attention mechanisms, i.e., the position attention module (PAM) and the channel attention (CA) module, are also employed in this local–global pyramid building feature extractor to adaptively select and emphasize some building features with high semantic responses. Furthermore, an effective cross-task transfer learning strategy (CTLS) is exploited to guide the proposed LGPNet to focus on the buildings, which further enhances the model's ability to represent the building features. The major contributions of this article are summarized as follows.

1) The LGPNet is established, which employs an LFPM and a GSPM to consider buildings of different scales from the global and local perspectives, respectively. This can avoid missing buildings of different scales and shapes, thus ameliorating the performance of the model for BCD effectively.
2) An effective CTLS is introduced in this article, by which the proposed LGPNet is able to effectively focus on buildings to weaken the false detections caused by other ground targets, thereby enhancing the generalization ability of the model for BCD.
3) The proposed LGPNet is comprehensively tested on two sets of open and challenging datasets (WHUCD [40] and LEVIR [43] with aerial and Google Earth sensors, respectively). The impressive experimental results have demonstrated that the proposed LGPNet can achieve the best detection results and outperform eight state-of-the-art comparison methods in terms of quantitative metrics and visual performances.

This article is organized as follows. In Section II, the related work is introduced. The proposed LGPNet is described in detail in Section III. Section IV presents and analyzes the experimental results. Section V concludes this article.

II. Related Works

In recent years, three techniques are becoming more popular, including pyramid structure, attention mechanism, and transfer learning. In this section, these three techniques are briefly reviewed in the application of remote sensing image or nature image processing.

A. Pyramid Structure

Pyramid structures have been devised for semantic segmentation in nature image processing [44]–[46]. In the early stage, many end-to-end semantic segmentation networks have been widely used, such as FCNs [28], U-Net [32], and Segnet [47]. However, these methods require a larger reception field to obtain more global category clues for multiscale objects. In this context, the pyramid structure is leveraged for expanding the reception field to obtain robust features of different objects with multiple scales. For example, [44] Chen *et al.* introduced atrous spatial pyramid pooling (ASPP) to acquire the features with multiple scales; Zhao *et al.* [45] deployed a pyramid scene parsing network (PSPNet) by aggregating features of the spatial pyramid with different scales in consideration of the global prior representation. In [46], a pyramid attention network (PANet) was devised to extract the high-level features that are combined with the low-level features to provide accurate semantic and locating information. Inspired by these methods, the pyramid structure is exploited in this article to excavate representative features for the buildings of various scales and shapes from local and global perspectives, respectively.

B. Attention Mechanism

Recent advancements in attention mechanisms have shown great potential to emphasize the meaningful information and weaken the undesirable information for extracting representative features in computer vision [48]–[52] and remote sensing image processing [11], [43], [53]. For these approaches, the

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attention mechanisms can be summarized into the following two categories.

1) *CA:* It is performed through a channel relationship matrix to weight the deep features from the channel dimension to enhance important channels and mitigate unexpected channels for obtaining discriminative features [48], [51], [53]. Nowadays, CA has been commonly utilized in remote sensing image CDs. For example, in [46], squeeze-and-excitation networks (SENets) [48] were combined with global pooling to construct a global coattention mechanism, which can achieve the high-level features to weight the low-level features for reinforcing the change information. Chen *et al.* [43] exploited the self-attention mechanism to establish CA for CD.
2) *Spatial Attention (SA):* It has been extensively applied to the field of computer vision and verified to be effective in increasing the ability of neural networks to capture more representative features [50]–[52]. In fact, SA indicates that the location relationship matrix is used to weigh the features of different locations from a spatial perspective, thereby highlighting the worthy regions and weakening the worthless regions. In other words, SA is able to mine the long-range dependencies of different locations for capturing representative features. Currently, many methods based on SA have been developed for CD. STANet [42] employs a self-attention mechanism to model the spatial–temporal relationship for CD; a difference-enhancement dense-attention CNN based on upsampling SA and CA was designed for optical image CD [54].

Given the advantages of the attention mechanism, it has been deployed as a regular module in a neural network to capture more representative features by considering the long-range dependencies of different channels and locations. Driven by this, we exploit the widely used attention mechanism to provide a robust feature extraction in the proposed LGPNet. The details will be provided in Section III-B.

C. Transfer Learning

Transfer learning means that the knowledge of related tasks can be transferred to the target data through a pretrained model on the source data (or an existing model) to help the generalization performance of the model on the target data [55], [56]. In recent years, this technique has been frequently applied in remote sensing CD. For instance, Zhan *et al.* [57] integrated bilinear CNNs based on transfer learning with object change analysis for unsupervised CD. A hyperspectral multiclass CD approach was promoted in [58], which transfers training samples of the source image for the classification of the target image, and then, the multiclass change map can be generated by the postclassification comparison. Other approaches based on transfer learning are available in [31] and [59]. In view of the fact that transfer learning has the capability to transfer relevant knowledge from the source data to the target data, a CTLS is introduced in this article. CTLS can transfer the knowledge of various buildings to enable the proposed LGPNet to pay attention to buildings more competently.

III. Proposed Local–Global Pyramid Network

In this section, an overview of the proposed LGPNet is first presented to express a whole framework of this model in Section III-A. To strengthen the building feature representation ability of the LGPNet, we then introduce the local–global pyramid building feature extractor, which is composed of two parts, i.e., the general attention modules [including the PAM and the CA module (CAM)] and the local–global pyramid modules (including the LFPM and the global spatial pyramid module). Following this, the details of each submodule are illustrated in Sections III-B and III-C, respectively. Finally, the CTLS is interpreted in Section III-D.

A. Overview of the Local–Global Pyramid Network

As shown in Fig. 2(a), the proposed LGPNet is a Siamese-based fully convolution network with an end-to-end structure, which is composed of the local–global pyramid building feature extractor with a weight-shared strategy and a simple feature fusion operation. The purpose of the local–global pyramid building feature extractor is to excavate deep representative features, so as to adapt to various buildings in the input image scene. Subsequently, the feature maps of the bitemporal images are generated and concatenated, which are fed into a convolution block for feature fusion then. Finally, the BCM can be acquired by the sigmoid classifier.

The structure of the local–global pyramid building feature extractor is a key structure in our proposed LGPNet as it can deliver the discriminative building feature representation. In the local–global pyramid building feature extractor, the encoder–decoder structure is built to effectively capture the deep features and recover the output scale of the feature extractor same as the scale of the input image, as shown in Fig. 2(b). Concretely, the structure of the encoder is consistent with the conventional U-Net [32] encoder, while the decoder is based on the U-Net decoder except that the CAM is employed at the end of each convolution block. That is because the decoder with the CAM can emphasize the significant feature maps that are helpful for restoring the building features during the decoding process. In addition to the encoder–decoder structure, the PAM is deployed at the top of the encoder to take the long-range dependencies between each building in the whole scene into consideration. In particular, we design two parallel branches after the PAM, including the LFPM branch and the GSPM branch. These two branches can enhance the representative building feature extraction effectively from local and global perspectives, respectively. Afterward, the feature maps of the two branches are concatenated and fed into the decoder to obtain the final feature map of the input image. The details of these modules in the local–global pyramid building feature extractor are provided in Sections III-B and III-C.

B. General Attention Modules

1) Position Attention Module: SA based on the nonlocal self-attention operation has been applied successfully in remote sensing image CD [42], [43]. These studies have demonstrated that SA based on the nonlocal self-attention

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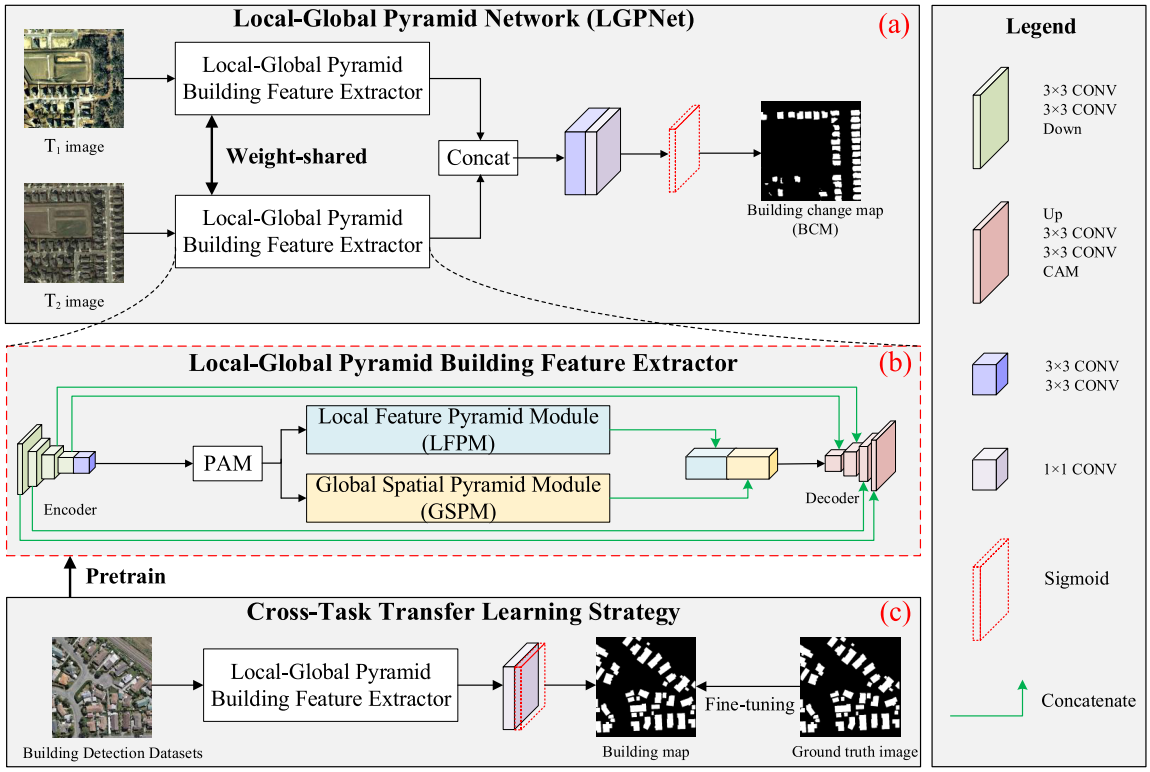


Fig. 2. Overview of the proposed LGPNet. (a) Architecture of the proposed LGPNet. (b) Local–global pyramid building feature extractor. (c) Schematic of the CTLS. (Notations: down refers to maxpool with stride is 2; up refers to upsampling operation by transpose convolution; PAM indicates the position attention module; CAM indicates the channel attention module; and LFPM and GSPM represent the local feature pyramid module and the global spatial pyramid module, respectively.)

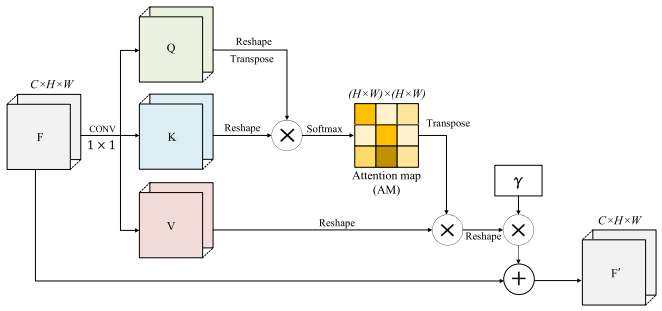


Fig. 3. Structure of the PAM.

operation can construct spatial long-range dependencies [42]. In other words, this mechanism can associate with each position of a single sequence to compute the representation of every position in the sequence [60]. In the task of BCD, the long-range dependencies between different buildings are beneficial for locating and identifying the buildings as the fact that various and discrete buildings typically exist in the remote sensing image scene. Considering the fact described above, a named PAM based on the self-attention mechanism in [52] is utilized in our local–global pyramid building feature extractor. The calculation details of the PAM are presented as follows.

As illustrated in Fig. 3, the feature $F \in \mathbb{R}^{C\times H\times W}$ is obtained by the encoder, where $H \times W$ is the size of the feature F and C indicates the number of channels in the feature F. In this context, the PAM can be calculated as the following steps. First, similar to the self-attention mechanism, we feed the feature F into a convolution layer with the 1×1 kernel size to acquire three new feature vectors, i.e., the query (Q), the key (K), and the value (V), where $\{Q, K, V\} \in \mathbb{R}^{C\times H\times W}$. Q and K are then reshaped to $\mathbb{R}^{C\times N}$, where $N = H \times W$ indicates the number of pixels. On this basis, matrix multiplication between the transposes of Q and K is first computed, and then, a softmax operation is performed to calculate the attention map (AM)

$$\mathrm{AM} = \mathrm{Softmax}\left(Q^T \times K\right) \tag{1}$$

where $\mathrm{AM} \in \mathbb{R}^{N\times N}$ and each element of AM, i.e., $\mathrm{AM}(i, j)$, can measure the correlation between the positions i and j. In addition, $(\cdot)^T$ denotes the transpose operation. The value of $\mathrm{AM}(i, j)$ will be greater if the correlation between the two positions is higher. Afterward, we reshape V to $\mathbb{R}^{C\times N}$ and perform a matrix multiplication between V and the transpose of AM, the result of which is reshaped to $\mathbb{R}^{C\times H\times W}$. Finally, the result is multiplied by a scale parameter γ and then summed with the feature F element-by-element to acquire the final output $F' \in \mathbb{R}^{C\times H\times W}$

$$F' = \gamma\left(\mathrm{AM}^T \times V\right) + F \tag{2}$$

where γ is a learnable scale parameter for the dynamic allocation of weights. Therefore, each position of F' is actually the weighted sum of the features at all positions and the original feature F. In other words, each position of F' can be considered as the relationship with all the other positions.

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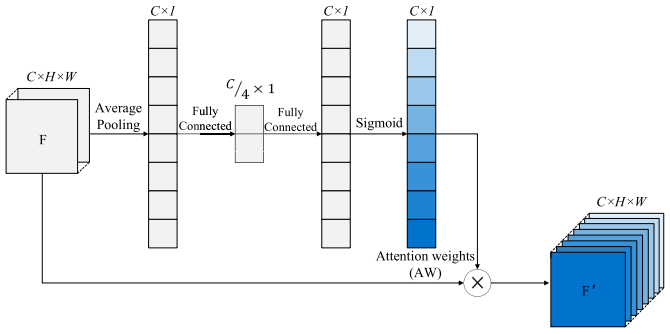


Fig. 4. Structure of the CAM.

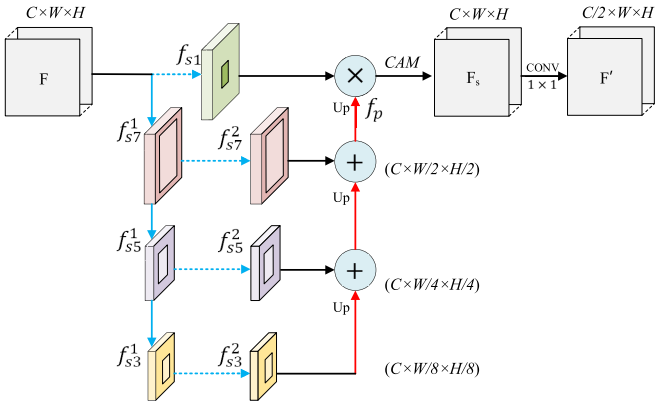


Fig. 5. Structure of the LFPM. The blue solid and dashed lines indicate convolution with the strides of 2 and 1, respectively, and the red line indicates the upsampling operation by bilinear interpolation.

According to this characteristic, long-range dependencies between buildings at different locations can be built for discriminative feature representation in our local–global building feature extractor.

2) Channel Attention Module: The CA mechanism has also been extensively used in CD [43], [61]. Similar to the SA mechanism, CA can be used to emphasize some significant feature channels as different feature channels have different semantic responses to the target of the interest [61]. In this fashion, some feature channels will be assigned with greater weights if the feature channels respond more significantly to the target of interest. In our local–global building feature extractor, the CAM in [62] is employed to emphasize feature channels that show high response to buildings.

The structure of the CAM is shown in Fig. 4. We assume that the feature vector is $F \in \mathbb{R}^{C\times H\times W}$, where C, H, and W denote the number of channels, height, and width of the feature F, respectively. The feature F can be unfolded as $F = \{f_1, f_2, f_3, \ldots, f_C\}$, where $f_n \in \mathbb{R}^{H\times W}$ is the slice of feature F and $n \in [1, C]$. First, adaptive average pooling is employed to f_n to generate a channelwise feature vector $fv = \{fv_1, fv_2, fv_3, \ldots, fv_C\}$ with $fv \in \mathbb{R}^C$. Then, two fully connected layers are executed to build channelwise dependencies. The result of the above fully connected layer is then mapped to [0, 1] through a sigmoid operation

$$\mathrm{AW} = \varphi(fv, \overline{W}) = \sigma(\Theta_2(\delta(\Theta_1(fv, W_1)), W_2)) \tag{3}$$

where $\mathrm{AW} = \{\mathrm{aw}_1, \mathrm{aw}_2, \ldots, \mathrm{aw}_C\}$ denotes the attention weights for the channelwise feature vector fv, $\overline{W}$ denotes the parameters regarding the above operations, $\Theta(\cdot)$ refers to the fully connected layer, $\sigma(\cdot)$ refers to the ReLU function, and $\delta(\cdot)$ refers to sigmoid operation. Finally, the output $F' \in \mathbb{R}^{C\times H\times W}$ of CAM can be acquired by the following formula:

$$F' = \mathrm{AW} \cdot F. \tag{4}$$

Hence, according to formula (4), CAM actually aims to calculate the weight vector that is used to weight the channelwise feature slices to enhance the feature slices, which has high semantic responses to buildings in the building feature extractor.

C. Local–Global Pyramid Modules

It has been empirically discovered that the local reception field is inadequate to locate and identify the targets of different shapes and scales precisely, especially very large or small targets [11], [63]. Moreover, the local reception field is also insufficient to capture the relationship between the buildings and other objects over the whole scene. In contrast, the global features can acquire global contextual information to alleviate effectively the limitations of using local features only. Focusing on this motivation, we devise an LFPM and a GSPM in the local–global pyramid building feature extractor. Among them, LFPM can provide more detailed local pixelwise localization information from a local perspective, while GSPM can capture more global contextual information from a global perspective. Meanwhile, LFPM and GSPM, respectively, incorporate features at different pyramid scales to reinforce the feature representation ability for buildings of different scales and shapes. Hence, the features of LFPM and GSPM are aggregated for fusing local and global features to capture a more robust building feature representation. In this subsection, the structural details of LFPM and GSPM are illustrated as follows.

1) Local Feature Pyramid Module: The structure of the LFPM is depicted in Fig. 5. We assume that the feature $F \in \mathbb{R}^{C\times H\times W}$ denotes the output of the PAM, where $H \times W$ is the size of the feature F, whereas C refers to the number of channels of the feature F. Then, the feature F is fed into LFPM. First, to better capture local location information of various buildings, a three-level local feature pyramid is consecutively deployed in LFPM. It can be presented as $f_{s7}^1 \in \mathbb{R}^{C\times(H/2)\times(W/2)}$, $f_{s5}^1 \in \mathbb{R}^{C\times(H/4)\times(W/4)}$, and $f_{s3}^1 \in \mathbb{R}^{C\times(H/8)\times(W/8)}$, which are obtained based on the feature F through a convolution with the stride of 2 and the convolution kernel sizes of 7×7, 5×5, and 3×3, respectively. Then, f_{s7}^2, f_{s5}^2, and f_{s5}^2 are acquired by convolution with the same kernel size but a stride of 1 to enhance the feature maps of the corresponding pyramid level. In this case, a large kernel size will not take a large amount of computation as the size of the feature F is small. Following this, the features of each pyramid level are merged layer-by-layer through summation to generate the pyramid feature $f_p \in \mathbb{R}^{C\times H\times W}$

$$f_p = \Psi\left(\Psi\left(\Psi\left(f_{s3}^2\right) + f_{s5}^2\right) + f_{s7}^2\right) \tag{5}$$

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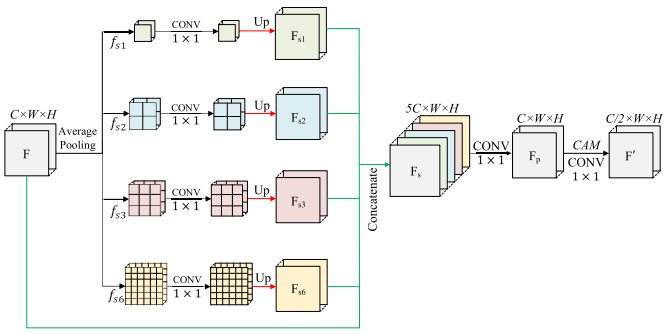


Fig. 6. Structure of the GSPM. Red lines denote the upsampling operation by bilinear interpolation. Green lines denote the concatenate operation.

where $\Psi(\cdot)$ denotes the upsampling operation by bilinear interpolation. Hence, this local feature pyramid is actually formed to a U-shape architecture. Afterward, the feature f_{s1} that is generated based on the feature F by a 1×1 convolution is multiplied pixelwise by the pyramid feature f_p. In this context, the original and pyramid features are aggregated, and CAM is employed to generate the feature F_s

$$F_s = \mathrm{CAM}\left(f_{s1} \cdot f_p\right) \tag{6}$$

where $\mathrm{CAM}(\cdot)$ indicates the CAM. Finally, a 1×1 convolution is applied to the feature F_s to generate the final output $F' \in \mathbb{R}^{(C/2)\times H\times W}$ of the LFPM. In this way, the local-detailed localization information can be captured by aggregating original and local pyramid features.

2) Global Spatial Pyramid Module: Previous semantic segmentation models, such as ASPP [44] and PSPNet [45], have verified that the use of spatial pyramid pooling at different scales can effectively capture the global contextual information. Considering the fact described above, similar to PSPNet [45], our GSPM is built through spatial pyramid pooling, which is composed of parallel branches based on four scales (1×1, 2×2, 3×3, and 6×6). As shown in Fig. 6, the feature $F \in \mathbb{R}^{C\times H\times W}$ is the input of the GSPM, where $H \times W$ is the size of the feature F, whereas C refers to the number of channels of the feature F. First, we adopt adaptive average pooling to the feature $F \in \mathbb{R}^{C\times H\times W}$ to generate the features of different pyramid scales ($f_{s1} \in \mathbb{R}^{C\times 1\times 1}$, $f_{s2} \in \mathbb{R}^{C\times 2\times 2}$, $f_{s3} \in \mathbb{R}^{C\times 3\times 3}$, and $f_{s6} \in \mathbb{R}^{C\times 6\times 6}$). Then, the global feature maps of the corresponding scale are acquired by using a 1×1 convolution and upsampling operation. Taking $f_{s1} \in \mathbb{R}^{C\times 1\times 1}$ as an example, it can be calculated by the following equation:

$$F_{s1} = \Psi\left(\xi\left(f_{s1}, \overline{\mathrm{W}}\right)\right) \tag{7}$$

where $\xi(\cdot)$ and $\overline{W}$ refer to a 1×1 convolution and the corresponding parameters, respectively, and $\Psi(\cdot)$ refers to the upsampling operation by bilinear interpolation. Similar to F_{s1}, F_{s2}, F_{s3}, and F_{s6} can also be obtained by this way. Thereafter, they are concatenated with the original feature F to form $F_s \in \mathbb{R}^{5C\times H\times W}$ for the combination of the features of each pyramid scale. This can be defined as

$$F_s = \{F \| F_{s1} \| F_{s2} \| F_{s3} \| F_{s6}\} \tag{8}$$

where $\|$ denotes the concatenation operation. Subsequently, to reduce the computational load, a 1×1 convolution is used to generate $F_p \in \mathbb{R}^{C\times H\times W}$. In this context, the CAM is then employed to F_p to select and stress significant channels in F_p. Finally, the final output $F' \in \mathbb{R}^{(C/2)\times H\times W}$ of the GSPM can be acquired after passing through a 1×1 convolution. Hence, in our local–global pyramid building feature extractor, GSPM can capture global semantic contextual clues effectively by spatial pyramid pooling with four different scales.

As described above, LFPM can provide detailed pixelwise localization information by aggregating local pyramid features layer by layer. In contrast, GSPM is better at capturing global contextual information by fusing global spatial pyramid features. Overall, they both deploy pyramid structures to cope with buildings of various shapes and scales. Moreover, LFPM and GSPM can excavate complementary information as the fact that the features are generated from local and global perspectives, respectively. Concretely, on the one hand, LFPM complements GSPM with detailed pixelwise location information to relieve the unexpected interference triggered by GSPM. On the other hand, GSPM provides global contextual clues to LFPM. Therefore, local–global pyramid modules based on LFPM and GSPM are employed in our local–global pyramid building feature extractor to capture more robust building feature representation from local and global perspectives, respectively.

D. Cross-Task Transfer Learning Strategy

It is acknowledged that sufficient data are essential for training a supervised deep neural network. However, the real situation often fails to meet expectations due to the fact that labeling data are labor-intensive and expensive. To overcome this issue, many studies have shown that transfer learning is a practical technology that can not only accelerate model convergence but also migrate relevant knowledge to improve the generalization ability of the model [55], [56]. In the BCD task, the performance of BCD is typically limited by the two factors as follows.

1) Different from building extraction, for BCD tasks, the number of changed buildings is usually much less than the total number of buildings, which leads to the difficulty of obtaining a robust model using only the BCD dataset.
2) In general, relative changes always exist in the bitemporal remote sensing images, while we only concentrate on changes of buildings in BCD. However, other land surface objects of various types are inevitable in the whole scene, such as grass, tree, road, bridge, vehicles, and parking areas. In this context, complex land cover targets may lead to false detections in BCM, as shown in the fourth and fifth rows with the yellow rectangle of Fig. 1.

Therefore, benefiting from transfer learning characteristics, a CTLS is introduced in the proposed LGPNet, and our CTLS is achieved by the following steps. First, to make the model better focus on buildings, several publicly available datasets for building detection tasks are collected for transferring

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Fig. 7. Bitemporal images and GT map examples from the training set of (a) WHUCD [41] and (b) LEVIR [42] BCD datasets. The first row denotes the T_1-time image. The second row denotes the T_2-time image. The third row denotes the GT map.

building-related knowledge. Although these building detection datasets are captured by different sensors and resolutions, the properties regarding the buildings are alike, such as the shape, scope, and distribution. To reach this aim, we then additionally deployed a 1×1 convolution and a sigmoid layer at the end of the local–global building feature extractor, as shown in Fig. 2(c). With this architecture, our local–global feature extractor can be pretrained based on these publicly available building detection datasets. Finally, it is initialized to a weight-shared local–global building feature extractor in the proposed LGPNet. Through the above implementation, our CTLS can effectively enhance the ability of the proposed LGPNet to recognize buildings and, thus, weaken the effect of complex background information on detection performance.

We assume that Y is the ground-truth (GT) map and $\bar{Y}$ is the predicted building map or BCM. Basically, our CTLS and LGPNet are trained to minimize the pixelwise binary cross-entropy (BCE) loss function, which is defined as

$$L(Y, \bar{Y}) = -\sum_{i=1}^{H \times W} y_i \log(\bar{y}_i) + (1 - y_i) \log(1 - \bar{y}_i) \quad (9)$$

where $H \times W$ denotes the size of the image. During the training period, CTLS first performs with the BCE loss based on a publicly available building detection dataset to pretrain the local–global building feature extractor in our proposed LGPNet. According to this, the entire LGPNet is then fine-tuned on the basis of a BCD dataset through the BCE loss. Finally, the excellent LGPNet model can be acquired for BCD in this way.

IV. Experiments and Analysis

To investigate the effectiveness of the proposed LGPNet, two publicly challenging BCD datasets, namely, WHUCD [40] and LEVIR [42], are employed in the experiments. In addition, three publicly building detection datasets are applied to execute our CTLS for pretraining our local–global building feature extractor. In this section, the following contents will be introduced. First, these datasets are depicted in detail. Second, evaluation metrics and several state-of-the-art CD methods are presented. Third, the implementation details of these methods are provided. Afterward, a series of ablation studies are deployed and discussed. Finally, experimental results based on two BCD datasets are compared and analyzed.

A. Dataset Descriptions

1) Building Change Detection Datasets: In the experiments, two datasets are used to test the performance of the proposed LGPNet. The details of these datasets are given as follows.

1) *WHUCD Dataset (Aerial Image):* As shown by some examples in Fig. 7(a), Ji *et al.* created an open BCD dataset, namely, WHUCD dataset, which is freely available in [41]. This dataset mainly depicts the building changes of Christchurch, New Zealand, where a 6.3 magnitude earthquake has occurred in February 2011. The WHUCD dataset is composed of two aerial images that were collected at the same location in 2012 and 2016. The dataset covers 20.5 km^2, with a size of $32\,507 \times 15\,354$ and a spatial resolution of 0.075 m. The BCD task based on the WHUCD dataset has the following challenges.
   a) Different building changes exist, including disappeared buildings and reconstructed buildings, which have various color, shapes, scales, and scope. There is a challenge to adapt to different forms of buildings and mine robust building features for a model.
   b) Except for the buildings, the whole remote sensing scene is covered by complex ground objects, such as grass, tree, road, bridge, vehicles, and parking areas. Hence, these ground objects are easily confused with buildings, again; they may also cause many false detected pixels in the BCM. In the first experiment, the WHUCD dataset is applied to verify the performance of the proposed LGPNet. To adapt to the GPU memory, we sequentially cropped the WHUCD dataset into 870 tiles with 512×512 pixels, which are divided into a training set (599 tiles) and a testing set (271 tiles).

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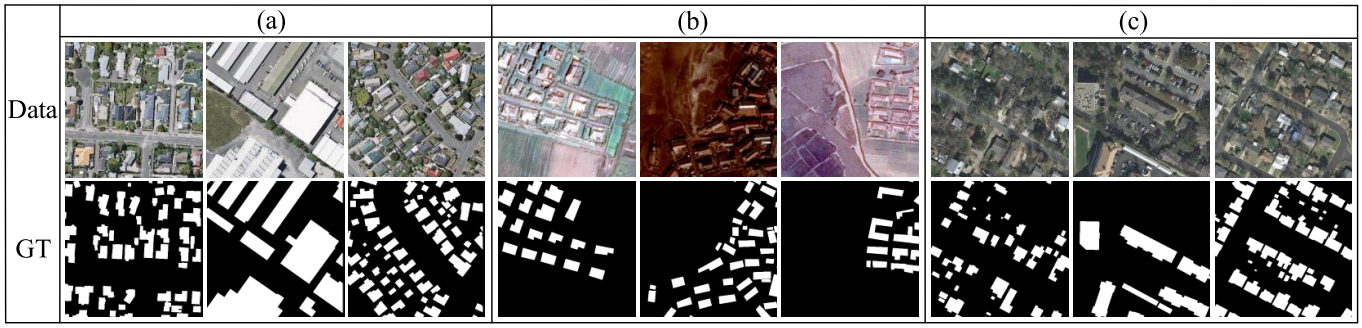


Fig. 8. Image and GT map examples from the training set of (a) WHU [41], (b) East Asia [41], and (c) Inria [64] building detection datasets. The first row denotes the building image. The second row denotes the GT map.

2) *LEVIR Dataset (Google Earth Image):* An open dataset called LEVIR has been developed in [42] for BCD, as shown by some examples in Fig. 7(b). The LEVIR dataset contains 637 image pairs that were collected by Google Earth API, Texas, USA, from 2002 to 2018. Among them, 445 image pairs are defined as a training set, while 192 image pairs are defined as a testing set. Each image pair has the size of 1024×1024 and a very high spatial resolution of 0.5 m. The LEVIR dataset has the following characteristics.
   a) Seasonal and illumination changes are obvious in bitemporal images, which is a challenging condition for BCD as it causes irrelevant changes easily in BCM.
   b) Various types of buildings are covered in the LEVIR dataset, such as villa residences, apartments, garages, and warehouses, which have different shapes, scales, and sizes as well. In the second experiment, the LEVIR dataset is employed to further test the effectiveness of the proposed LGPNet. This dataset was cropped into 2548 tiles with 512×512 pixels, including a training set (1780 image pairs) and a testing set (768 image pairs).

2) Building Detection Datasets: In our experiments, three different building detection datasets are adopted to perform CTLS to pretrain the local–global building feature extractor in the proposed LGPNet. These datasets are mainly designed to evaluate the influence of different datasets on our CTLS. The details of these datasets are given as follows.

1) *WHU Dataset (Aerial Image):* Same as the WHUCD dataset, the WHU dataset has also been proposed in [41], which is used for the building detection task. The WHU dataset contains 8188 aerial images with 512×512 pixels and a spatial resolution of 0.075 m, which were captured in Christchurch, New Zealand, as shown in Fig. 8(a). These images are originally divided into a training set (4736 images), a validation set (1036 images), and a testing set (2416 images).
2) *East Asia Dataset (Satellite Image):* Except for the WHU dataset, a satellite dataset was also provided in [41], namely, the East Asia dataset [see Fig. 8(b)]. This dataset covers 550 km^2 in East Asia, with a ground resolution of 2.7 m. Here, we select the part with buildings in this dataset, and it contains 3153 images in the training set and 903 images in the testing set.
3) *Inria Dataset (Aerial Image):* As shown in Fig. 8(c), the Inria dataset is freely available for building detection in [64]. This dataset contains 180 aerial images and covers 405 km^2, which were collected in a series of different regions, including Austin, Chicago, Kitsap County. Each image has a size of 5000×5000 and a spatial resolution of 0.3 m. In our experiments, we selected 37 images in Austin from Inria [64]. These images were cropped sequentially into 2997 tiles with 512×512 pixels, which were divided into a training set (2025 tiles) and a testing set (891 tiles).

B. Evaluation Metrics and Comparative Methods

1) Evaluation Metrics: In this article, seven widely used evaluation metrics, including overall accuracy (OA), changed intersection over union (CIoU), unchanged intersection over union (UCIoU), mean intersection over union (mIoU), precision (PRE), recall (REC), and F1-score (F1), are employed to quantitatively evaluate the performance of the proposed LGPNet approach from different perspectives [41]–[43], [65]. From this, true positive (TP) denotes the number of pixels correctly detected as changed buildings; true negative (TN) denotes the number of pixels correctly detected as unchanged buildings or nonbuilding; false positive (FP) denotes the number of pixels falsely detected as changed buildings; and false negative (FN) denotes the number of missing pixels that are not detected as changed buildings. Based on this, these evaluation metrics can be calculated as follows:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{10}$$

$$\mathrm{CIoU} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN}} \tag{11}$$

$$\mathrm{UCIoU} = \frac{\mathrm{TN}}{\mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{12}$$

$$\mathrm{mIoU} = \frac{1}{2} \times (\mathrm{CIoU} + \mathrm{UCIoU}) \tag{13}$$

$$\mathrm{PRE} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{14}$$

$$\mathrm{REC} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{15}$$

$$\mathrm{F1} = \frac{2 \times \mathrm{PRE} \times \mathrm{REC}}{\mathrm{PRE} + \mathrm{REC}}. \tag{16}$$

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2) Comparative Methods: To verify the effectiveness of the proposed LGPNet approach, eight related and state-of-the-art methods are selected to compare with our proposed LGPNet in the experiments. Here, each method is presented as follows.

(1) *FC_EF [35]:* The FC_EF model was proposed based on U-Net, which performs an early fusion strategy by concatenation of the input bitemporal images [35].
(2) *FC_Siam_Conc [35]:* It was promoted in [35], which shares weights based on two same encoding layers of U-Net. In this model, two skip connections are concatenated in the decoding part for later fusion.
(3) *FC_Siam_Diff [35]:* Different from FC_Siam_Conc, FC_Siam_Diff was also developed in [35], which concatenates the absolute value of the difference of two skip connections instead.
(4) *SegNet [47]:* It is a classic pixelwise semantic segmentation network, which is built based on an end-to-end encoder–decoder architecture [47].
(5) *PSPNet [45]:* It was proposed in [45]. PSPNet is similar to our proposed LGPNet model as it exploits the multiscale pyramid pooling operation to consider the global context information. Hence, PSPNet is here selected to compare with the proposed LGPNet.
(6) *PANet [46]:* A PANet was designed in [46], which provides abundant detailed information and global semantic guidance by a feature pyramid attention and global attention upsampling, respectively. In view of the characteristics of PANet, it is chosen for comparison in the experiments.
(7) *SiU-Net [43]:* It was devised for the WHU dataset [43]. This model consists of two branches with weight sharing, and then, the original image and their downsampling counterparts are used as the input of two branches, respectively. This model is able to mitigate the challenges resulting from the buildings of different scales and shapes to some extent. Therefore, the comparison of SiU-Net with the proposed LGPNet is meaningful.
(8) *STANet [42]:* In the literature [42], the LEVIR BCD dataset was created. Meanwhile, an STANet was also developed to benchmark the LEVIR dataset, which aggregates a spatial pyramid based on spatial–temporal attention. In the experiments, it is necessary to compare STANet with the proposed LGPNet.

C. Implementation Details

In the experiments, we performed the proposed LGPNet on the Pytorch platform by using a single NVIDIA GTX 1080Ti GPU (11-GB video memory) and a single NVIDIA Tesla P40 GPU (24-GB video memory). CTLS was first implemented on a single NVIDIA GTX 1080Ti for pretraining local–global building feature extractor, and the proposed LGPNet was then trained on a single NVIDIA Tesla P40. During the training period, the Adam optimizer with a weight delay of $1e-5$ was employed to minimize the loss. The base learning rate was set to $1e-4$, and a multistep learning rate decay strategy with the gamma of 0.9 was adopted to update the learning rate. In all experiments, the batch size of the training set was set to 4. Two experimental datasets have the same hyperparameter settings.

TABLE I
QUANTITATIVE EVALUATION ACCURACY (IN %) FOR ABLATION STUDY OF CTLS ON THE WHUCD DATASET

Training strategy			OA	mIoU	F1
Without CTLS			96.09	82.20	81.43
With CTLS	CTLS datasets	WHU	98.21	91.08	91.39
		Inria	98.39	91.86	92.19
		East Asia	98.17	90.92	91.22

In addition to the above settings, data augmentation is extensively used in deep learning for enhancing the robustness of the model [41], [63]. Hence, in all experiments, including our proposed LGPNet and other comparative methods, data augmentation was also adopted for more robust models. At the training phase, the input bitemporal images with a size of 512×512 pixels were randomly transformed through three flip operations (horizontal flip, vertical flip, and horizontal–vertical flip). Based on these deployments, the experiments are performed for verifying the performance and advantages of our proposed LGPNet.

D. Ablation Study on WHUCD Dataset

To mitigate the influence of the complex background information, a CTLS is introduced to pay attention to the buildings for the BCD tasks. Moreover, to better capture the robust building features, especially the buildings with very large or small scale, we introduce the general attention modules and devise the local–global pyramid modules in the proposed LGPNet. To verify the effectiveness of these components, two ablation studies are deployed on the WHUCD dataset as follows. Besides, this section exploits three comprehensive metrics (OA, mIoU, and F1) to quantitatively evaluate the results of ablation experiments.

1) Ablation Study for CTLS: The ablation study that we focus on is the effect of CTLS in the proposed LGPNet approach. In this ablation experiment, the proposed LGPNet without CTLS was first executed. Then, three pretraining datasets (WHU, Inria, and East Asia) were employed to perform CTLS for pretraining the local–global pyramid building feature extractor in the proposed LGPNet, respectively. The effectiveness of CTLS is comprehensively validated by comparing the nonCTLS-based LGPNet with the CTLS-based LGPNet on three different pretraining datasets.

According to the above target, the ablation experimental results were acquired on the WHUCD dataset. The quantitative evaluation for the whole dataset is presented in Table I, and some representative examples of the visualization results are shown in Fig. 9. For these representative examples, the CTLS-based method obtains more accurate building detection maps. For example, in Fig. 9, the yellow rectangle in the first row depicts a region actually without building changes. However, this area contains some ground objects that share similar color and shape with the buildings that are vulnerably detected as the buildings. In this context, the results with CTLS on each pretraining dataset obviously reduce the false detected pixels in the BCMs. Moreover, the results with CTLS have

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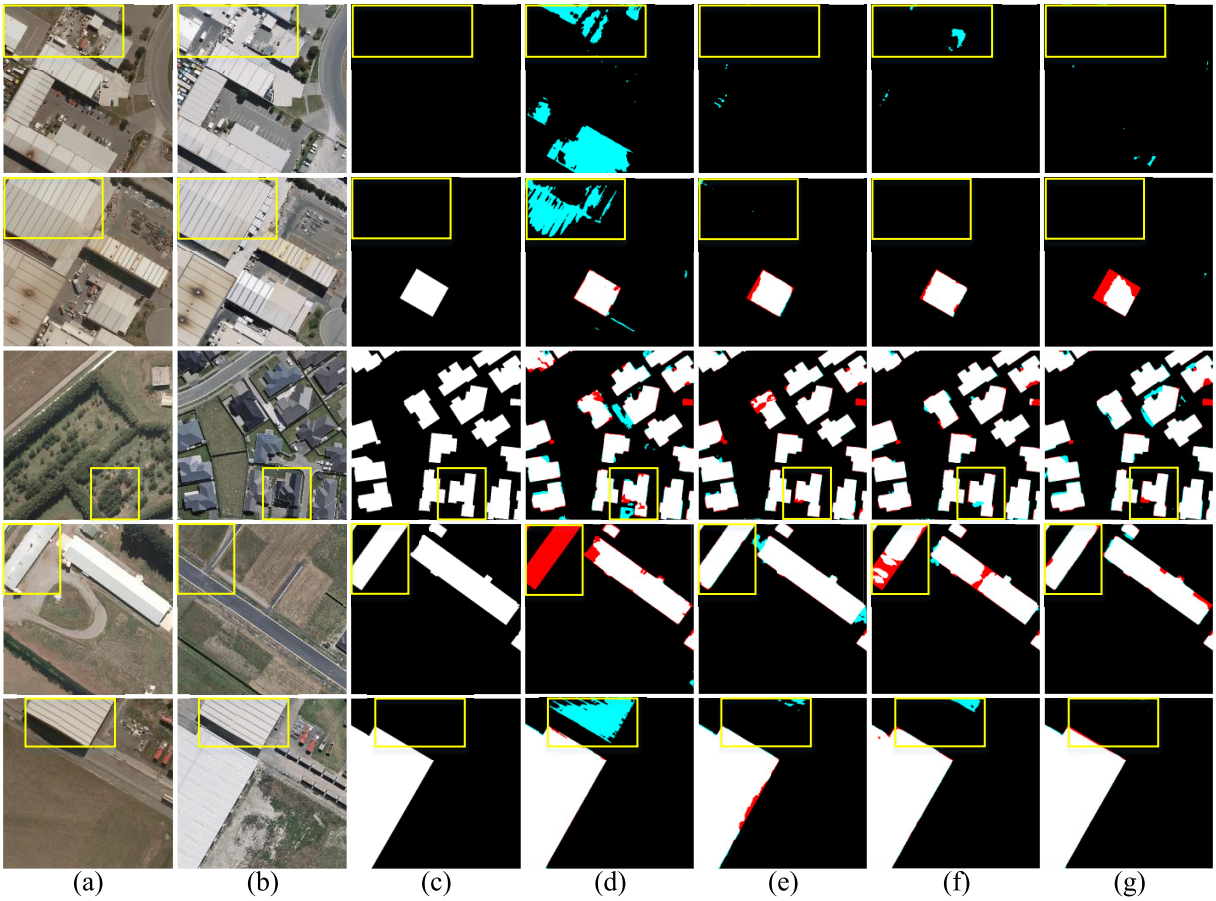


Fig. 9. Representative examples of visualization results of ablation study for CTLS on the WHUCD dataset: (a) T_1-time image, (b) T_2-time image, (c) GT map, (d) without CTLS, (e) with CTLS on the WHU dataset, (f) with CTLS on the Inria dataset, and (g) with CTLS on the East Asia dataset. (Notations: red, cyan, white, and black denote missed detection pixels, false detection pixels, correct detection changed pixels, and correct detection unchanged pixels, respectively.)

TABLE II
QUANTITATIVE EVALUATION ACCURACY (IN %) FOR ABLATION STUDY OF EACH SUBMODULE ON THE WHUCD DATASET

Submodel Name	Backbone (M1)	CAM (M2)	PAM (M3)	LFPM (M4)	GSPM (M5)	OA	mIoU	F1
M1	✓					97.62	88.64	88.83
M1+M2	✓	✓				97.79	88.87	88.99
M1+M2+M3	✓	✓	✓			97.77	89.25	89.48
M1+M2+M3+M4	✓	✓	✓	✓		98.03	90.26	90.53
M1+M2+M3+M5	✓	✓	✓		✓	97.92	89.74	89.97
M1+M2+M3+M4+M5	✓	✓	✓	✓	✓	98.39	91.86	92.19

some boost on the buildings of various shapes and scales, as presented in the third and fourth rows in Fig. 9. In addition, our method with CTLS can provide significant improvement for the quantitative evaluation of the whole WHUCD dataset. Notably, the CTLS on different pretraining datasets can always reach our motivational expectations.

2) Ablation Study for Each Submodule: In order to explore the effect of each submodule, the proposed LGPNet architecture is decomposed into five submodules: backbone, CAM, PAM, LFPM, and GSPM. Among them, the backbone indicates that only the original U-Net [32] is used as the building feature extractor instead of the proposed local–global pyramid building feature extractor. In this ablation experiment, each submodule is gradually combined and compared for the BCD task.

As shown in Table II, the results of combining different submodules were evaluated, respectively, for the WHUCD dataset. When the backbone network and the general attention modules (CAM and PAM) are combined for BCD, a performance boost in terms of mIoU and F1 (0.61% and 0.65%) can be observed, while the improvement of OA is slight. Hereafter, the network with the LFPM or GSPM can obtain obvious promotion. Among them, the network combined with GSPM can achieve the accuracy of 97.92%, 89.74%, and 89.97% for

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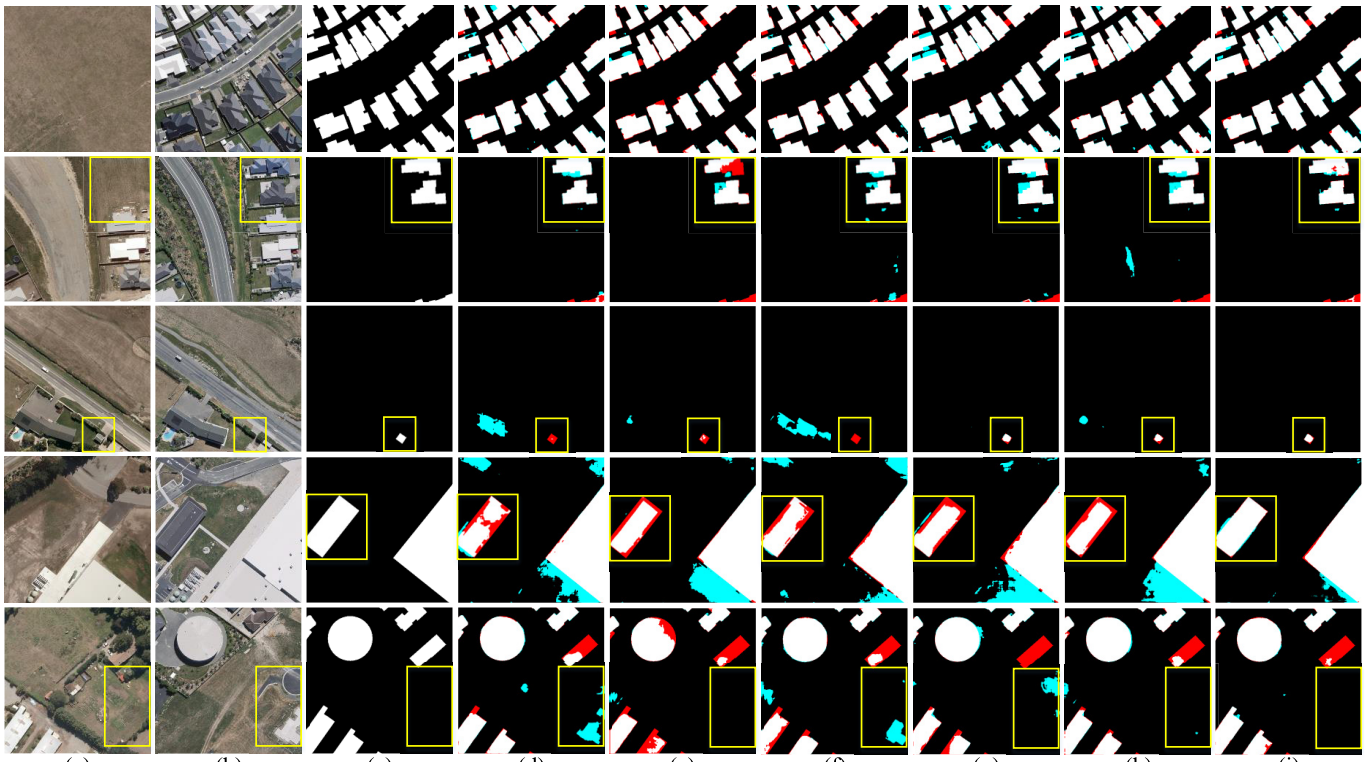


Fig. 10. Representative examples of the visualization results of ablation study for each submodule on the WHUCD dataset: (a) T_1-time image, (b) T_2-time image, (c) GT map, (d) backbone, (e) CAM, (f) PAM, (g) LFPM, (h) GSPM, and (i) proposed LGPNet. (Notations: red, cyan, white, and black denote missed detection pixels, false detection pixels, correct detection changed pixels, and correct detection unchanged pixels, respectively.)

OA, mIoU, and F1, respectively, while the improvement of 0.26%, 1.01%, and 1.05% can be reached in terms of OA, mIoU, and F1, respectively, for the network combined with LFPM. Finally, all submodules are combined as the proposed LGPNet, and it reaches the best accuracy (98.39%, 91.86%, and 92.19%) for the WHUCD dataset. These quantitative results demonstrate that the combination of our LFPM and GSPM can effectively improve the performance of the model. Some representative examples of the visual results in this ablation study for each submodule also provide the same conclusions as to the quantitative results, as presented in Fig. 10. It is worth noting that the yellow rectangular in Fig. 10 demonstrates more clearly that the combination of LFPM and GSPM in LGPNet significantly enhances the model's ability to represent the features of various buildings.

Overall, by the preceding ablation experiments, two characteristics can be found.

(1) Our CTLS has been verified that it is useful for the BCD tasks to productively alleviate false detected pixels in the BCM triggered by complicated background information. Moreover, CTLS can also enhance the robustness of the building feature representation of the proposed LGPNet to some certain extent.

(2) Compared with the backbone network, both the CAM and the PAM improve the accuracy of the model. However, the accuracy improvement is limited. Besides, by introducing a single LFPM or a single GSPM, the performance of the model can be further increased. When the LFPM and the GSPM are combined, local location information and global semantic contextual information are utilized and fused adequately, and the comprehensive accuracy of the network is increased obviously.

E. Experimental Comparison and Analysis

To evaluate the superiority of the proposed LGPNet, the proposed model is compared with other eight existing methods, including FC_EF [35], FC_Siam_Conc [35], FC_Siam_Diff [35], SegNet [47], PSPNet [45], PANet [46], SiU-Net [41], and STANet [42] on WHUCD and LEVIR datasets. These approaches are recently proposed with well-applied multilevel integration and context aggregation strategies, such as the pyramid structure-based model and the attention mechanism-based model for semantic segmentation tasks or CD tasks. Therefore, these methods were conducted for experimental comparison and analysis in this section.

1) Results of WHUCD Dataset: The BCMs were acquired by various methods on the WHUCD dataset, and some representative examples are shown in Fig. 11. Among them, the results of the proposed LGPNet were obtained with CTLS based on the Inria dataset. From the visual observations, with respect to the GT map, the proposed LGPNet can visibly provide less noise compared with other methods. Furthermore, the adaptability of the proposed LGPNet is better than all

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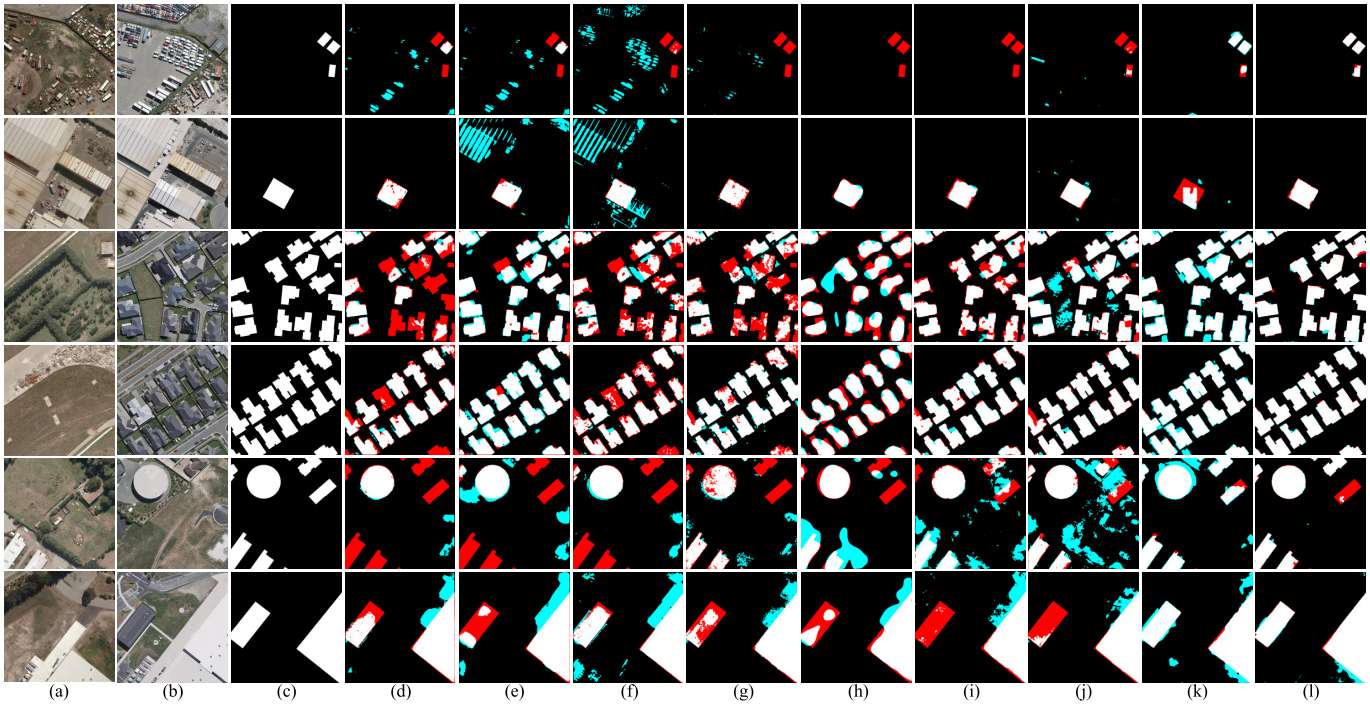


Fig. 11. Representative examples of the visualization results of various methods on the WHUCD dataset: (a) T_1-time image, (b) T_2-time image, (c) GT map, (d) FC_EF [35], (e) FC_Siam_Conc [35], (f) FC_Siam_Diff [35], (g) SegNet [47], (h) PSPNet [45], (i) PANet [46], (j) SiU-Net [41], (k) STANet [42], and (l) proposed LGPNet. (Notations: red, cyan, white, and black denote missed detection pixels, false detection pixels, correct detection changed pixels, and correct detection unchanged pixels, respectively.)

TABLE III
QUANTITATIVE COMPARISON RESULTS (IN %) OF VARIOUS METHODS APPLIED ON THE WHUCD DATASET

Methods	OA	CIoU	UCIoU	mIoU	PRE	REC	F1
FC_EF [35]	95.73	63.39	95.39	79.39	86.35	70.46	77.59
FC_Siam_Conc [35]	94.45	61.90	93.90	77.90	68.91	85.89	76.47
FC_Siam_Diff [35]	91.92	47.38	91.28	69.33	59.91	69.37	64.29
SegNet [47]	96.36	69.34	96.04	82.68	85.69	78.41	81.89
PSPNet [45]	96.04	65.91	95.71	80.81	87.14	73.01	79.45
PANet [46]	96.74	71.57	96.45	84.01	89.52	78.12	83.43
SiU-Net [41]	96.82	73.59	96.51	85.05	85.19	84.38	84.78
STANet [42]	96.96	76.40	96.63	86.52	80.42	**93.87**	86.62
Proposed LGPNet	**98.39**	**85.51**	**98.22**	**91.86**	**93.84**	90.59	**92.19**

the other eight comparison approaches in terms of bitemporal remote sensing images with complex urban backgrounds. For example, as presented in the first row of Fig. 11, the bitemporal images cover many objects that are similar to buildings. Most methods have difficulty detecting building changes in this scene, except STANet [42] and the proposed LGPNet. In addition, for the visual performance of these methods, although the state-of-the-art STANet [42] can also provide a more complete BCM, many false detected pixels are inevitable in BCM compared with the proposed LGPNet.

In addition to the visual results, quantitative comparison results of these methods are also presented in Table III, and the proposed LGPNet reaches the best accuracies (98.39%, 85.51%, 98.22%, 91.86%, 93.84%, and 92.19%) than those of the eight compared methods in terms of OA, CIoU, UCIoU, mIoU, PRE, and F1. Despite that STANet [42] can obtain the best accuracy of REC, it thickens the detected buildings, thereby leading to overdetection. These quantitative comparisons further strengthen the preceding conclusions, which means that the comprehensive superiority of the proposed LGPNet can be achieved for the WHUCD dataset.

2) Results of LEVIR Dataset: The BCMs were obtained by various methods on the LEVIR dataset, and some representative examples are shown in Fig. 12. Among them, the results of the proposed LGPNet were generated with CTLS on the WHU dataset. Through observing Fig. 12, the proposed LGPNet can provide a clearer BCM for changed buildings with various shapes and scales. For instance, some small changed buildings are shown in the first and third rows of Fig. 12; many small changed buildings miss detection by most approaches except STANet [42] and our proposed LGPNet. Moreover, some large or irregular changed buildings are presented in the fifth and

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TABLE IV
QUANTITATIVE COMPARISON RESULTS (IN %) OF VARIOUS METHODS APPLIED ON THE LEVIR DATASET

Methods	OA	CIoU	UCIoU	mIoU	PRE	REC	F1
FC_EF [35]	97.87	60.89	97.80	79.35	83.81	69.01	75.69
FC_Siam_Conc [35]	98.33	67.09	98.27	82.68	92.04	71.23	80.31
FC_Siam_Diff [35]	98.23	65.88	98.16	82.02	89.42	71.45	79.43
SegNet [47]	97.47	48.61	97.41	73.01	94.74	49.96	65.42
PSPNet [45]	97.08	50.25	96.99	73.62	73.45	61.41	66.89
PANet [46]	98.00	64.49	97.93	81.21	81.34	75.69	78.42
SiU-Net [41]	98.18	63.31	98.12	80.71	**95.01**	65.49	77.53
STANet [42]	98.24	71.11	98.16	84.64	76.98	**90.32**	83.12
Proposed LGPNet	**99.16**	**83.63**	**99.12**	**91.38**	92.87	89.38	**91.09**

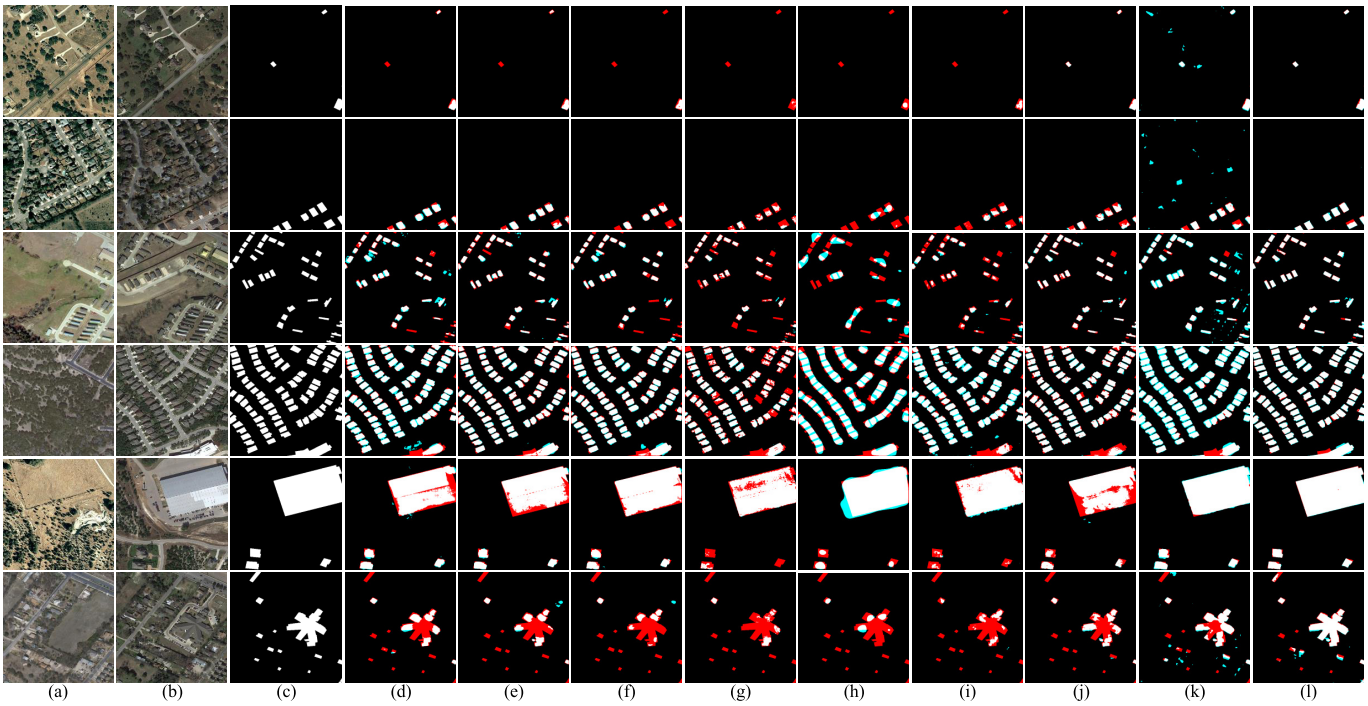


Fig. 12. Representative examples of visualization results of various methods on the LEVIR dataset: (a) T_1-time image, (b) T_2-time image, (c) GT map, (d) FC_EF [35], (e) FC_Siam_Conc [35], (f) FC_Siam_Diff [35], (g) SegNet [47], (h) PSPNet [45], (i) PANet [46], (j) SiU-Net [41], (k) STANet [42], and (l) proposed LGPNet. (Notations: red, cyan, white, and black denote missed detection pixels, false detection pixels, correct detection changed pixels, and correct detection unchanged pixels, respectively.)

sixth rows of Fig. 12; only our proposed LGPNet can achieve relatively a more complete BCM.

Apart from visual results, quantitative comparison results of various methods are presented in Table IV. Compared with the quantitative results of different methods, the proposed LGPNet can achieve the best accuracies (99.16%, 83.63%, 99.12%, 91.38%, and 91.09%) in terms of OA, CIoU, UCIoU, mIoU, and F1. Significant improvements are acquired by the proposed LGPNet. For example, compared with SiU-Net [41], the improvements reached by the proposed LGPNet are approximately 0.98%, 10.68%, and 13.56% for three comprehensive evaluation metrics (OA, mIoU, and F1); for state-of-the-art STANet [42], the improvements of 0.98%, 10.68%, and 13.56% are obtained for OA, mIoU, and F1. In addition, we can discover that SiU-Net [41] provides the best accuracy of PRE, while STANet gets the best accuracy of REC, which is because SiU-Net [41] thins out the detected buildings, which results in more missed pixels in the BCM. Furthermore, STANet [42] thickens the detected buildings, which results in more false pixels in the BCM. Overall, our proposed LGPNet can obtain state-of-the-art performance for the LEVIR dataset.

In summary, compared with the eight selected methods, the proposed LGPNet provides the best performance for both the WHUCD dataset and the LEVIR dataset. Considering the fact above, two characteristics should be noticed.

(1) The proposed LGPNet has proved that it can provide a better generalization for the buildings of different scales and shapes. That is because the local location information and the global semantic information can be fully fused by exploiting our LFPM and GSPM in the local–global building feature extractor; thereby, the

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representation capability of robust building features can be increased.

(2) The proposed LGPNet is better at discriminating the changed buildings with various colors and shapes from the nonbuilding background region in the images. The reason is that our CTLS can not only facilitate architectural feature extraction to a certain extent but also effectively alleviate the interference of complex background, which shares similar characteristics (colors, shapes, and so on) with buildings, and thus, significantly reduce false detected pixels in BCM. Hence, compared with other selected methods, the proposed LGPNet is more competitive for BCD.

V. Conclusion and Future Research

In this article, an LGPNet has been proposed for BCD. In the proposed LGPNet, a local–global building feature extractor with Siamese structure is devised based on the encoder–decoder architecture, which deploys an LFPM and a GSPM that work in parallel to capture the local location information and the global semantic contextual information, respectively. Afterward, LFPM and GSPM are combined and fused fully to enhance feature discrimination of various buildings in a complementary manner. In addition, a CTLS is introduced to the training process to alleviate the interference of nonbuilding areas in the scene, while building-related knowledge is migrated from the datasets of other tasks to further enhance the robustness of the features. The proposed LGPNet has been tested comprehensively based on the WHUCD dataset [41] (Aerial) and the LEVIR dataset [42] (Google Earth), which are available and challenging BCD datasets. A series of ablation studies have confirmed the effectiveness of the proposed LGPNet. Moreover, comparison experiments have demonstrated that the proposed LGPNet can provide superior performance compared with all the other eight selected methods.

In the future research, three aspects will be considered: 1) we will concentrate on the interpretation of the combination of the local detailed information and global contextual information; 2) we will promote the method to a weakly supervised or an unsupervised BCD; and 3) the method will be extended to general CD.

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References

[1] D. Lu, P. Mausel, E. Brondízio, and E. Moran, "Change detection techniques," *Int. J. Remote Sens.*, vol. 25, no. 12, pp. 2365–2401, 2004.

[2] G. Chen, G. J. Hay, L. M. T. Carvalho, and M. A. Wulder, "Object-based change detection," *Int. J. Remote Sens.*, vol. 33, no. 14, pp. 4434–4457, 2012.

[3] M. Bouziani, K. Goïta, and D.-C. He, "Automatic change detection of buildings in urban environment from very high spatial resolution images using existing geodatabase and prior knowledge," *ISPRS J. Photogramm. Remote Sens.*, vol. 65, no. 1, pp. 143–153, Jan. 2010.

[4] M. M. Rathore, A. Ahmad, A. Paul, and S. Rho, "Urban planning and building smart cities based on the Internet of Things using big data analytics," *Comput. Netw.*, vol. 101, no. 4, pp. 63–80, Jun. 2016.

[5] J.-F. Zhang, L.-L. Xie, and X.-X. Tao, "Change detection of earthquake-damaged buildings on remote sensing image and its application in seismic disaster assessment," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, vol. 4, Jul. 2003, pp. 2436–2438.

[6] P. Washaya, T. Balz, and B. Mohamadi, "Coherence change-detection with Sentinel-1 for natural and anthropogenic disaster monitoring in urban areas," *Remote Sens.*, vol. 10, no. 7, p. 1026, Jun. 2018.

[7] J. Tian, S. Cui, and P. Reinartz, "Building change detection based on satellite stereo imagery and digital surface models," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 1, pp. 406–417, Jan. 2013.

[8] Y. Tang, X. Huang, and L. Zhang, "Fault-tolerant building change detection from urban high-resolution remote sensing imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 10, no. 5, pp. 1060–1064, Sep. 2013.

[9] L. ZhiYong, T. Liu, J. A. Benediktsson, and N. Falco, "Land cover change detection techniques: Very-high-resolution optical images: A review," *IEEE Geosci. Remote Sens. Mag.*, early access, Jul. 8, 2021, doi: 10.1109/MGRS.2021.3088865.

[10] Y. Liu, C. Pang, Z. Zhan, X. Zhang, and X. Yang, "Building change detection for remote sensing images using a dual-task constrained deep Siamese convolutional network model," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 811–815, May 2021.

[11] H. Jiang, X. Hu, K. Li, J. Zhang, J. Gong, and M. Zhang, "PGA-SiamNet: Pyramid feature-based attention-guided Siamese network for remote sensing orthoimagery building change detection," *Remote Sens.*, vol. 12, no. 3, p. 484, Feb. 2020.

[12] X. Huang, L. Zhang, and T. Zhu, "Building change detection from multitemporal high-resolution remotely sensed images based on a morphological building index," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 1, pp. 105–115, Jan. 2014.

[13] Z. Y. Lv, T. F. Liu, P. Zhang, J. A. Benediktsson, T. Lei, and X. Zhang, "Novel adaptive histogram trend similarity approach for land cover change detection by using bitemporal very-high-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 9554–9574, Dec. 2019.

[14] Y. Tian, M. Hao, and H. Zhang, "Unsupervised change detection using spectrum-trend and shape similarity measure," *Remote Sens.*, vol. 12, no. 21, p. 3606, Nov. 2020.

[15] Z. Lv, T. Liu, C. Shi, and J. A. Benediktsson, "Local histogram-based analysis for detecting land cover change using VHR remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 7, pp. 1284–1287, Jul. 2021.

[16] X. X. Zhu *et al.*, "Deep learning in remote sensing: A comprehensive review and list of resources," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 8–36, Dec. 2017.

[17] L. Ma, Y. Liu, X. Zhang, Y. Ye, G. Yin, and B. A. Johnson, "Deep learning in remote sensing applications: A meta-analysis and review," *ISPRS J. Photogramm. Remote Sens.*, vol. 152, pp. 166–177, Jun. 2019.

[18] D. Wen *et al.*, "Change detection from very-high-spatial-resolution optical remote sensing images: Methods, applications, and future directions," *IEEE Geosci. Remote Sens. Mag.*, early access, Apr. 5, 2021, doi: 10.1109/MGRS.2021.3063465.

[19] Z. Shao, P. Tang, Z. Wang, N. Saleem, S. Yam, and C. Sommai, "BRRNet: A fully convolutional neural network for automatic building extraction from high-resolution remote sensing images," *Remote Sens.*, vol. 12, no. 6, p. 1050, Mar. 2020.

[20] L. Zhang, M. Lan, J. Zhang, and D. Tao, "Stagewise unsupervised domain adaptation with adversarial self-training for road segmentation of remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, early access, Aug. 18, 2021, doi: 10.1109/TGRS.2021.3104032.

[21] M. Lan, Y. Zhang, L. Zhang, and B. Du, "Global context based automatic road segmentation via dilated convolutional neural network," *Inf. Sci.*, vol. 535, pp. 156–171, Oct. 2020.

[22] H. Gao, Y. Miao, X. Cao, and C. Li, "Densely connected multiscale attention network for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 2563–2576, Feb. 2021. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/9345335

[23] X. Jia, Y. Peng, J. Li, B. Ge, Y. Xin, and S. Liu, "Dual-complementary convolution network for remote-sensing image denoising," *IEEE Geosci. Remote Sens. Lett.*, early access, Aug. 12, 2021, doi: 10.1109/LGRS.2021.3101851.

[24] K. Jiang, Z. Wang, P. Yi, G. Wang, T. Lu, and J. Jiang, "Edge-enhanced GAN for remote sensing image superresolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5799–5812, Aug. 2019.

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[25] L. Khelifi and M. Mignotte, "Deep learning for change detection in remote sensing images: Comprehensive review and meta-analysis," *IEEE Access*, vol. 8, pp. 126385–126400, 2020.
[26] W. Shi, M. Zhang, R. Zhang, S. Chen, and Z. Zhan, "Change detection based on artificial intelligence: State-of-the-art and challenges," *Remote Sens.*, vol. 12, no. 10, p. 1688, May 2020.
[27] M. Gong, J. Zhao, J. Liu, Q. Miao, and L. Jiao, "Change detection in synthetic aperture radar images based on deep neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 27, no. 1, pp. 125–138, Jan. 2015.
[28] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.
[29] E. Maggiori, Y. Tarabalka, G. Charpiat, and P. Alliez, "Fully convolutional neural networks for remote sensing image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 5071–5074.
[30] Z. Zhong, J. Li, W. Cui, and H. Jiang, "Fully convolutional networks for building and road extraction: Preliminary results," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 1591–1594.
[31] A. Song and J. Choi, "Fully convolutional networks with multiscale 3D filters and transfer learning for change detection in high spatial resolution satellite images," *Remote Sens.*, vol. 12, no. 5, p. 799, Mar. 2020.
[32] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent*. Berlin, Germany: Springer, Nov. 2015, pp. 234–241.
[33] Z. Lv, T. Liu, X. Kong, C. Shi, and J. A. Benediktsson, "Landslide inventory mapping with bitemporal aerial remote sensing images based on the dual-path fully convolutional network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 4575–4584, Mar. 2020. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/9036919
[34] D. Peng, Y. Zhang, and H. Guan, "End-to-end change detection for high resolution satellite images using improved UNet++," *Remote Sens.*, vol. 11, no. 11, p. 1382, 2019.
[35] R. C. Daudt, B. L. Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.
[36] L. Li, C. Wang, H. Zhang, B. Zhang, and F. Wu, "Urban building change detection in SAR images using combined differential image and residual U-Net network," *Remote Sens.*, vol. 11, no. 9, p. 1091, May 2019.
[37] W. Wiratama, J. Lee, and D. Sim, "Change detection on multispectral images based on feature-level U-Net," *IEEE Access*, vol. 8, pp. 12279–12289, 2020.
[38] S. Ji, Y. Shen, M. Lu, and Y. Zhang, "Building instance change detection from large-scale aerial images using convolutional neural networks and simulated samples," *Remote Sens.*, vol. 11, no. 11, p. 1343, Jun. 2019.
[39] S. Saha, F. Bovolo, and L. Bruzzone, "Building change detection in VHR SAR images via unsupervised deep transcoding," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 1917–1929, Mar. 2021.
[40] Y. Sun, X. Zhang, J. Huang, H. Wang, and Q. Xin, "Fine-grained building change detection from very high-spatial-resolution remote sensing images based on deep multitask learning," *IEEE Geosci. Remote Sens. Lett.*, early access, Sep. 7, Sep. 2020, doi: 10.1109/LGRS.2020.3018858.
[41] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019.
[42] H. Chen and Z. Shi, "A spatial-temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, 2020.
[43] J. Chen *et al.*, "DASNet: Dual attentive fully convolutional Siamese networks for change detection in high-resolution satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1194–1206, Nov. 2021. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/9259045
[44] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2017.
[45] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2881–2890.
[46] H. Li, P. Xiong, J. An, and L. Wang, "Pyramid attention network for semantic segmentation," 2018, *arXiv:1805.10180*.
[47] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder-decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017.
[48] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.
[49] T. Xiao, Y. Xu, K. Yang, J. Zhang, Y. Peng, and Z. Zhang, "The application of two-level attention models in deep convolutional neural network for fine-grained image classification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2015, pp. 842–850.
[50] F. Wang *et al.*, "Residual attention network for image classification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Apr. 2017, pp. 3156–3164.
[51] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, Sep. 2018, pp. 3–19.
[52] J. Fu *et al.*, "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3146–3154.
[53] L. Wang, J. Peng, and W. Sun, "Spatial–spectral squeeze-and-excitation residual network for hyperspectral image classification," *Remote Sens.*, vol. 11, no. 7, p. 884, Apr. 2019.
[54] X. Peng, R. Zhong, Z. Li, and Q. Li, "Optical remote sensing image change detection based on attention mechanism and image difference," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7296–7307, Sep. 2021.
[55] F. Zhuang *et al.*, "A comprehensive survey on transfer learning," *Proc. IEEE*, vol. 109, no. 1, pp. 43–76, Jul. 2020.
[56] C. Tan, F. Sun, T. Kong, W. Zhang, C. Yang, and C. Liu, "A survey on deep transfer learning," in *Proc. 27th Int. Conf. Artif. Neural Netw.* Berlin, Germany: Springer, Sep. 2018, pp. 270–279.
[57] T. Zhan, M. Gong, X. Jiang, and W. Zhao, "Transfer learning-based bilinear convolutional networks for unsupervised change detection," *IEEE Geosci. Remote Sens. Lett.*, early access, Apr. 8, 2021, doi: 10.1109/LGRS.2021.3070145.
[58] X. Tong *et al.*, "A novel approach for hyperspectral change detection based on uncertain area analysis and improved transfer learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 2056–2069, Apr. 2020. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/9080097
[59] S. Saha, Y. T. Solano-Correa, F. Bovolo, and L. Bruzzone, "Unsupervised deep transfer learning-based change detection for HR multispectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 856–860, May 2021.
[60] A. Vaswani *et al.*, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 6000–6010.
[61] K. Song and J. Jiang, "AGCDetNet:An attention-guided network for building change detection in high-resolution remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4816–4831, May 2021. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/9423508
[62] T. Zhao and X. Wu, "Pyramid feature attention network for saliency detection," in *Proc. CVPR*, Jun. 2019, pp. 3085–3094.
[63] H. Zhang, Y. Liao, H. Yang, G. Yang, and L. Zhang, "A local-global dual-stream network for building extraction from very-high-resolution remote sensing images," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Dec. 16, 2020, doi: 10.1109/TNNLS.2020.3041646.
[64] E. Maggiori, Y. Tarabalka, G. Charpiat, and P. Alliez, "Can semantic labeling methods generalize to any city? The inria aerial image labeling benchmark," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 3226–3229.
[65] Z. Lv, G. Li, Z. Jin, J. A. Benediktsson, and G. M. Foody, "Iterative training sample expansion to increase and balance the accuracy of land classification from VHR imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 139–150, Jan. 2021.
[66] F. Gao, J. Dong, B. Li, and Q. Xu, "Automatic change detection in synthetic aperture radar images based on PCANet," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 12, pp. 1792–1796, Dec. 2016.



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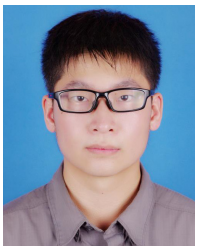
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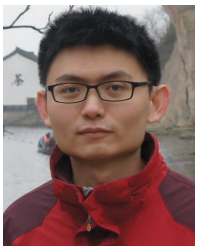
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